

Real Time Chat App

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Introduction

- Real-time chat applications allow instant communication between users.
- Messages are delivered without refreshing the page.
- Messaging apps like WhatsApp and Messenger are widely used today.
- These apps support text messaging, image sharing, and online status.
- This project uses the MERN stack and Socket.io for real-time communication.

Technologies Used

Backend Technologies

- **Node.js** – Server runtime
- **Express.js** – Backend framework
- **MongoDB** – Database
- **JWT** – Authentication

Frontend Technologies

- **React.js** – Frontend UI
- **Tailwind CSS** – UI styling

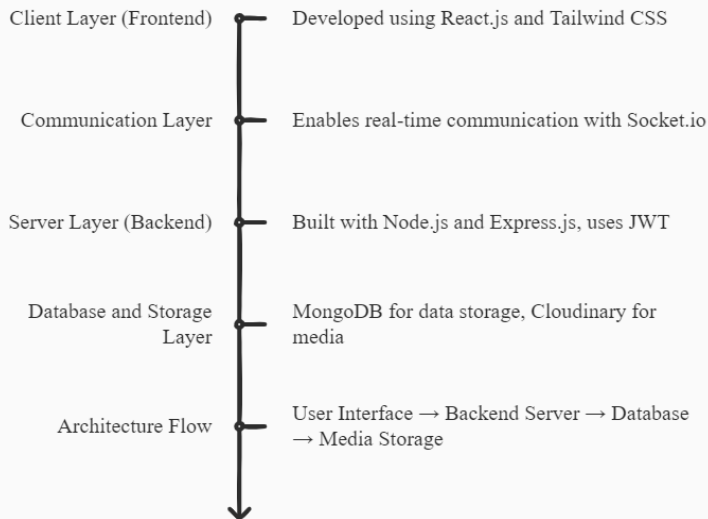
Real-Time Communication

- **Socket.io** – Enables instant messaging

Media Storage

- **Cloudinary** – Image storage and management

Real-Time Chat Application Architecture



Features of the Chat Application

1 User Signup and Login

- Users can create accounts and log in securely.

2 Real-time Messaging

- Messages are delivered instantly using Socket.io.

3 Image Sharing

- Users can share images within the chat.

4 Online / Offline Status

- Users can see whether others are online or offline.

5 Profile Editing

- Users can update profile information and images.

6 Unseen Message Notification

- Users receive notifications for unread messages.

7 Responsive UI

- The application works smoothly on mobile and desktop devices.

Backend Implementation

- Built using **Node.js** and **Express.js**
- REST APIs for authentication and messaging
- Passwords secured using **bcryptjs**
- **JWT** used for user authentication

Main API Endpoints

- /signup – User registration
- /login – User login
- /messages – Send / receive messages
- /profile – Manage user profile

Real-Time Communication (Socket.io)

Key Concept

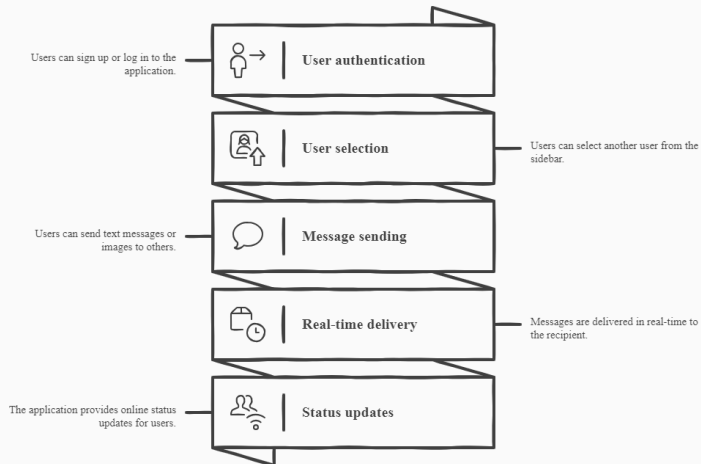
- Enables **bidirectional communication** between client and server
- Server keeps track of **online users**
- Messages are delivered **instantly** without page refresh

Socket Events

- `connect` – User connects to the server
- `new message` – Send and receive messages
- `disconnect` – User leaves the chat

Application Demo

App functionality



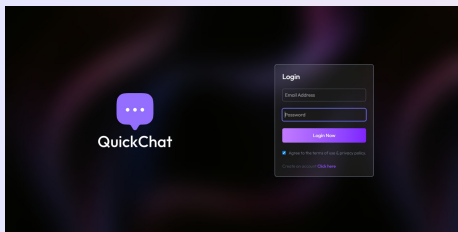
Authentication System

- **JWT (JSON Web Token)** is used for authentication
- Token is generated after successful login/signup
- Token is stored in **localStorage** on client side
- Each request includes token for verification
- **Middleware** is used to verify user identity
- Unauthorized users are blocked from accessing APIs
- Passwords are securely hashed using **bcryptjs**
- Ensures data security and user privacy

Login and Signup Pages

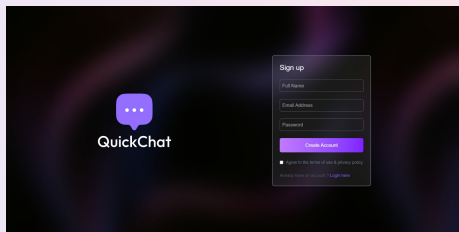
Login Page

- User enters email and password
- JWT token generated on success
- Redirected to chat interface



Signup Page

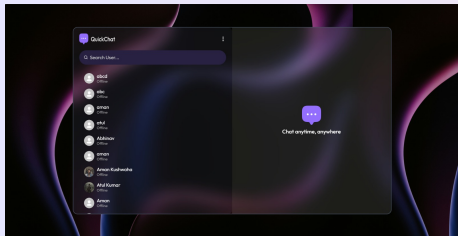
- User creates a new account
- Data stored in MongoDB
- Password is hashed securely



Home Page and Chat Interface

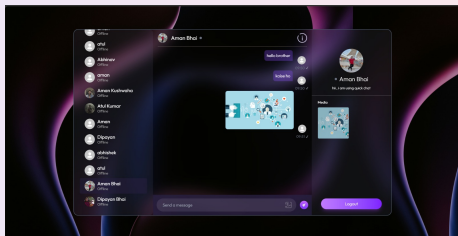
Home Page

- Displays user dashboard after login
- Shows contacts and recent chats
- Provides navigation to profile and settings



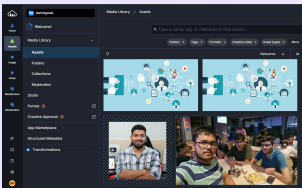
Chat Interface

- Real-time messaging using Socket.io
- Supports text and image messages
- Shows online/offline user status



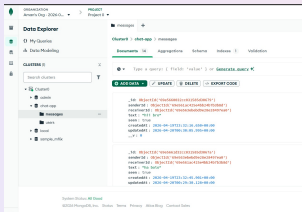
Data Storage Components

Cloudfinary



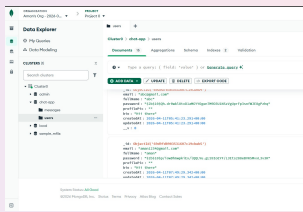
Stores images and media files securely

Message Database



Stores chat messages with timestamps

User Database



Stores user details and authentication data

Advantages

- **Fast Real-Time Communication**

Messages are delivered instantly between users.

- **Scalable Architecture**

MERN stack supports multiple users efficiently.

- **Secure Authentication**

JWT ensures safe and secure user login.

- **Cloud-Based Storage**

Images and media stored securely using Cloudinary.

Future Improvements

- Voice and video calls
- Group chats
- Message encryption
- Push notifications
- Mobile app version

Thank You!

Any Questions?